



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE(Advanced)
REVIEW TEST
04-04-2021

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A), I, LIVE-I(A) & LIVE-I]

PAPER-2

Time : 3 Hours

Maximum Marks : 198

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Please see the last page of this booklet for rest of the instructions

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

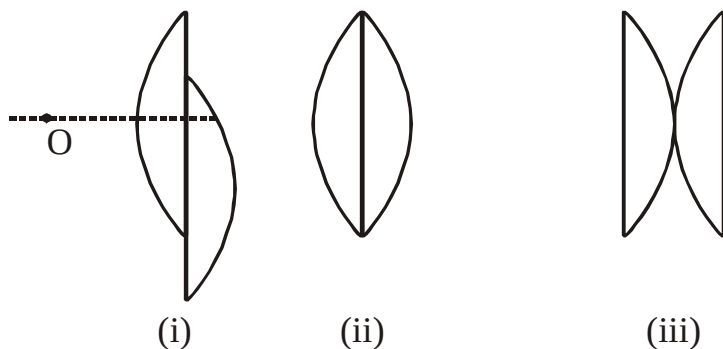
SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

-
1. A particle moves in a plane according to equation $\vec{r} = A \cos 2\omega t \hat{i} + \frac{A}{2} \cos(2\omega t) \hat{j}$. Motion of particle is :
- (A) Periodic (B) On a straight line
(C) On an ellipse (D) Oscillatory
-

Space for Rough Work

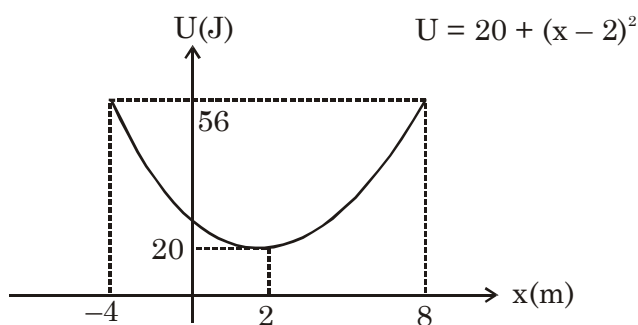
2. Which of the following functions represents the SHM ?
 (A) $x = \sin \omega t^2$ (B) $x = \sin^2 \omega t$
 (C) $x = \sin \omega t + 2 \cos \omega t$ (D) $x = k + \sin 2 \omega t$ ($k = \text{constant}$)
3. A particle is moving towards a fixed convex mirror along its principle axis. If v_o is speed of object and v_i is speed of image then :
 (A) $v_o > v_i$ if $|u| > |f|$ (B) $v_o < v_i$ if $|u| < |f|$
 (C) $v_o > v_i$ if $|u| < |f|$ (D) $v_o < v_i$ if $|u| > |f|$
4. If a symmetrically biconvex lens is cut into two halves and placed in different ways as shown in figure.



- (A) Two images will be formed in case (i)
 (B) Three images will be formed in case (i)
 (C) Ratio of focal lengths in case (ii) to (iii) is 2
 (D) Ratio of focal lengths in case (ii) to (iii) is 1

Space for Rough Work

5. For the refraction of light through a prism kept in air.
- (A) For every angle of deviation there are two angles of incidence
 - (B) For isosceles prism light inside it is always parallel to base when there is minimum deviation
 - (C) There are two angles of incidence for maximum deviation
 - (D) If refractive index of prism is increased then angle of minimum deviation will increase surely.
6. A particle moves under the action of a force whose corresponding potential energy is as shown. If particle has kinetic energy of 30 J at $x = 4$ m then :



- (A) Maximum value of KE is 56 J
- (B) Mechanical energy of system is 56 J
- (C) Mechanical energy of system is 54 J
- (D) Maximum value of KE is 34 J

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

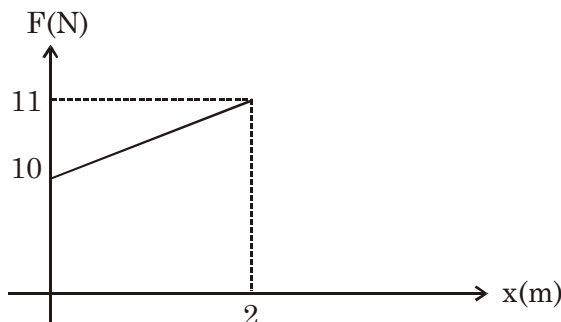
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

		+			
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	●
6	6	6	6	6	6
7	7	●	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

		-			
●	●	0	0	0	0
1	1	1	1	1	1
2	2	2	2	●	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	●	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

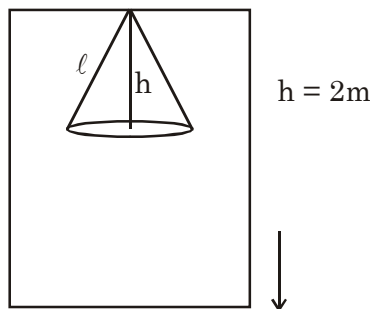
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. There is a small air bubble inside a glass sphere of refractive index $\frac{3}{2}$ and radius 20 cm. Bubble is at distance of 5 cm from centre. Find its apparent depth in cm from surface when viewed normally from the side where it is nearest to surface ?
2. A force F acts on a particle of $m = 1\text{kg}$ along x-axis whose variation is as shown. If initially particle was at rest then find it's speed at $x = 2\text{m}$ in m/s ?



Space for Rough Work

3. A force $\vec{F} = (3t\hat{i} + 4\hat{j})\text{N}$ acts on a body and displacement of body varies with time as $\vec{s} = (2t^2\hat{i} - 8\hat{j})\text{m}$ then work of force in 2 sec will be (in J)
4. A circular turn of radius $40\sqrt{3}\text{m}$ has angle of banking 30° . At what speed (in m/s) should a vehicle go so that no friction is used ?
5. A lift goes downward with constant retardation. Conical pendulum in the lift is revolving with time period of 2 sec. Find retardation of lift (m/s^2). Use $\pi^2 = 10$.



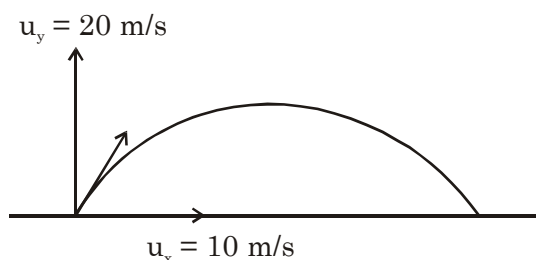
6. A body of mass 1 kg hangs at one end of a string of length 2m, other end of which is fixed. It is given some velocity at its lowest position so that string would just become slack, when it makes an angle of 60° with the upward drawn vertical line. Find tension (in N) in string at the point of projection ?

Space for Rough Work

SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

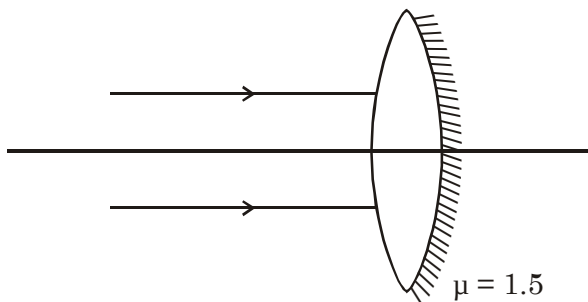
1. Two particle are performing SHM along same line about same mean position. Initially both are at their positive extremes. If time period of each particle is 24 sec. and difference of their amplitude is 12 cm then find minimum time (in sec) after which the separation between particles become 6 cm.
2. A stone is projected from ground as shown in figure at $t = 0$. Then find minimum time (in sec) after which magnitude of tangential and normal acceleration become same?



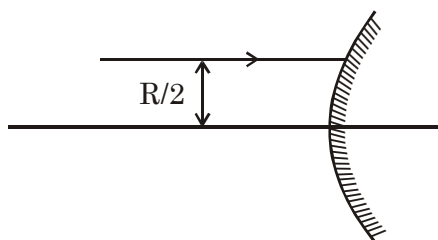
3. A block is placed on a platform performing SHM vertically with period of π sec. Minimum amplitude (in m) of motion at which block & platform will separate is A. Then fill value of 2A.

Space for Rough Work

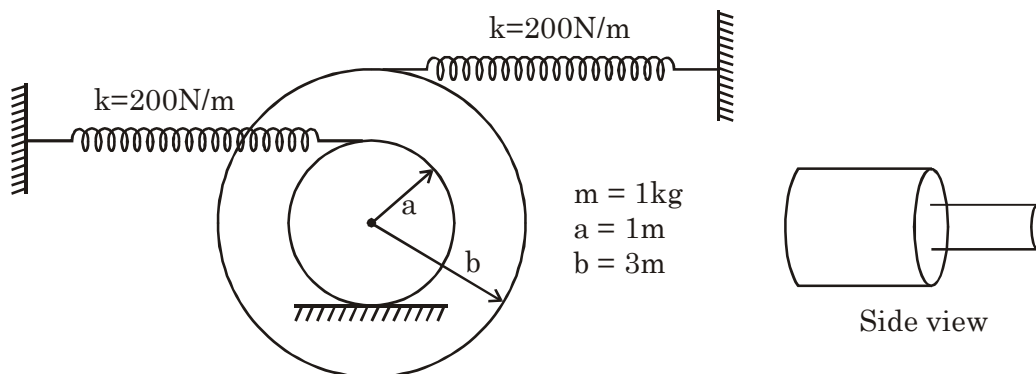
4. A parallel light beam falls on a lens which is silvered from back side as shown and both surfaces have radius 25 cm then distance (in cm) from lens where the beam converges on principle axis is x so fill value of $\frac{16x}{25}$.



5. Radius of curvature of mirror is $\sqrt{3}m$ then find distance of point on principle axis from centre of curvature where light ray appears to cut it after reflection from mirror.



6. If this body rolls purely then time period of its small oscillations is T . It's moment of inertia about CM is 9 kg.m^2 . Fill value of $\frac{40}{\pi}T$?



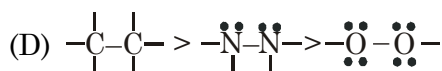
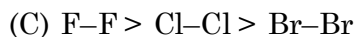
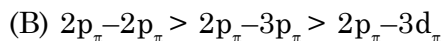
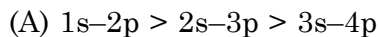
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Select **CORRECT** order of relative bond strength for given overlapping.

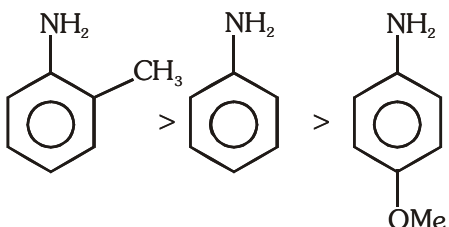


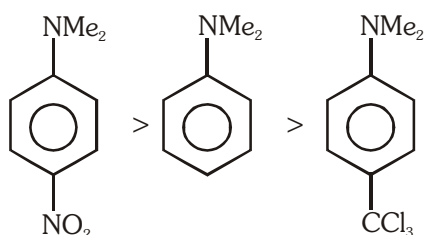
Space for Rough Work

2. Which of the following is / are correct order of basic nature.

(A) $\text{Me}_2\text{NH} > \text{Me} - \text{NH}_2 > \text{Me}_3\text{N}$ in aqueous solution.

(B) $\text{CH}_3 - \overset{\text{NH}}{\underset{\text{||}}{\text{C}}} - \text{NH}_2 > \text{CH}_3 - \text{CH}_2 - \text{NH}_2 > \text{C}_6\text{H}_5\text{NH}_2$

(C) 
 $\text{C}_6\text{H}_4(\text{NH}_2)\text{CH}_3 > \text{C}_6\text{H}_5\text{NH}_2 > \text{C}_6\text{H}_4(\text{NH}_2)\text{OMe}$

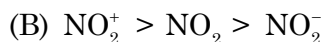
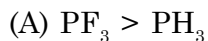
(D) 
 $\text{C}_6\text{H}_4(\text{NHMe}_2)\text{NO}_2 > \text{C}_6\text{H}_5\text{NMe}_2 > \text{C}_6\text{H}_4(\text{NHMe}_2)\text{CCl}_3$

3. Select the **CORRECT** statement(s) -

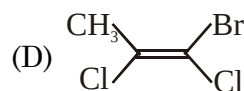
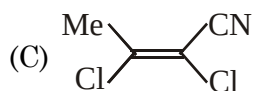
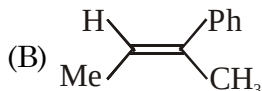
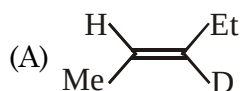
- (A) A crystalline solid possesses long range order while amorphous solid may have only short range order.
- (B) Crystalline solids have definite & characteristic heat of fusion and melting temperature while amorphous solids do not have definite heat of fusion and soften over a range of temperature
- (C) When polyurethane and benzoic acid are cut with sharp edge tool former has irregular cleavage surface while later has smooth surface.
- (D) Quartz (crystalline) is isotropic while quartz glass is anisotropic in nature.

Space for Rough Work

4. Select the **CORRECT** order of bond angle



5. Which of the following represent(s) E form



6. In a structure of ionic compound anions are present in hcp structure while cation occupy 2/3 part of octahedral voids. Select the correct statement(s)

(A) Fraction of total voids occupied by cation is 2/9

(B) Fraction of total voids occupied by cation is 2/3

(C) If anion is (A^{2-}), then cation can be (B^+)

(D) If anion is (A^{2-}), then cation can be (B^{3+})

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

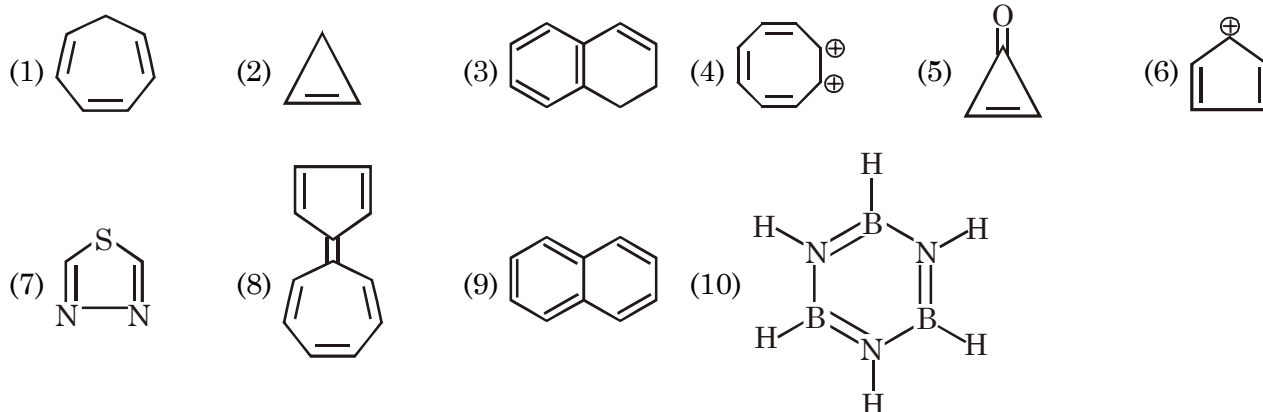
+					-				
0	0	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. How many of the following orders are correct against indicated properties.

- $\text{CaH}_2 > \text{SrH}_2 > \text{BaH}_2$ (Thermal stability)
- $\text{LiOH} < \text{NaOH} < \text{KOH}$ (Water solubility)
- $\text{NaCl} > \text{KCl} > \text{LiCl}$ (Melting point)
- $\text{AgCl} < \text{AgBr} < \text{AgI}$ (Intensity of colour)
- $\text{CaCl}_2 > \text{ZnCl}_2 > \text{KCl}$ (Covalent character)

2. How many of following compound are aromatic in nature



Space for Rough Work

3. When 0.2 g of acetic acid is added to 20 g of benzene, its freezing point decreases by 0.45°C .

If % dimerisation of acetic acid in benzene is 'X%'. Find value of $\left(\frac{X}{10}\right)$

K_f (Benzene) = $5 \text{ K}\cdot\text{kg mol}^{-1}$.

4. Find the sum of oxidation numbers of the central atom of oxyacids obtained by hydrolysis of following molecules :-

BrF_5 , SO_2Cl_2 , BCl_3

5. How many species is/are $-\text{M} / -\text{R}$ groups

(1) $-\text{OH}$

(2) $-\text{NO}_2$

(3) $-\text{CN}$

(4) $-\text{CH}_3$

(5) $-\text{O}^-$

(6) $-\text{NH}_3^+$

(7) $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OH} \end{array}$

(8) $-\text{CH}=\text{NH}$

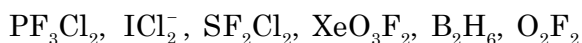
6. Oxygen gas is maintained at partial pressure of 0.81 atm above water sample. Henry's constant for dissolution of O_2 in water is 10^5 atm . If molality of O_2 in water at equilibrium is $(y \times 10^{-4})$ final value of 'y'

Space for Rough Work

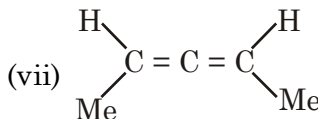
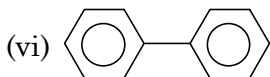
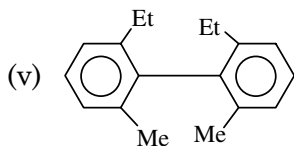
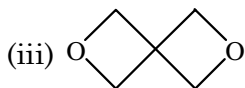
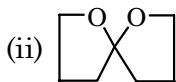
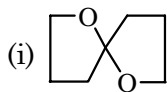
SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. How many of the following is/are polar species.



2. Select chiral molecules.

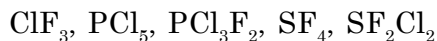


Space for Rough Work

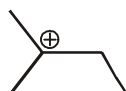
3. One litre of a sample of hard water contains 33.3 mg of CaCl_2 & 11.7 mg of NaCl. What is degree of hardness in terms of ppm of CaCO_3 (Given : Atomic mass of Ca = 40 , Na = 23, Cl = 35.5)

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

4. Among the following, find out total number of species in which axial bond length is greater than equatorial bond length.



5. Identify total number of hyper conjugating structures involving C–H bond in following carbocation?



6. For an aqueous solution of 0.1M $\text{Ba}(\text{NO}_3)_2$, osmotic pressure at 300K is found to be 6.0 atm. Percentage dissociation of $\text{Ba}(\text{NO}_3)_2$ in solution is. ($R = 0.08 \text{ atm-L/mole-K}$)

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 24)

- This section contains **SIX** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Let $f(x) = \cot^{-1}(\text{sgn}(x)) + \sin^{-1}(x - \{x\})$ which of the following is(are) correct ?
 (where $\{x\}$ is fractional part of x) (where $\text{sgn}(x)$ is signum of x)
 (A) Domain of $f(x)$ is $[-1, 2)$
 (B) $f(x)$ is an even function $\forall x \in [-1, 1]$
 (C) $f(x)$ is bounded

(D) Number of solution of the equation $f(x) = \frac{\pi}{2}$ is zero

2. $\lim_{x \rightarrow \infty} \sqrt{x} (\sqrt{x+1} - \sqrt{x})$ is equal to

(A) $\lim_{x \rightarrow 0} \frac{\ln(1+x) - x}{x^2}$

(B) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$

(C) $\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - 1}{x}$

(D) $\lim_{x \rightarrow 0} \frac{\sqrt{x}}{\sqrt{x + \sqrt{x^2 + 2x}}}$

Space for Rough Work

3. If $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = 2x - \sin x$ and $g(x) = \sqrt[3]{x}$, then
 (A) range of $g \circ f$ is $\mathbb{R}$ (B) $g \circ f$ is one-one
 (C) both f and g are one-one (D) both f and g are onto
4. Let $f(x) = \frac{1 - \cos\{x\}}{(x^4 + ax^3 + bx^2 + cx)^2}$. If $\ell = \lim_{x \rightarrow 1^+} f(x)$, $m = \lim_{x \rightarrow 2^+} f(x)$ and $n = \lim_{x \rightarrow 3^+} f(x)$, where ℓ , m and n are non-zero finite then (where $\{x\}$ denotes the fractional part of x)
 (A) $\lim_{x \rightarrow 0^+} f(x) = \frac{1}{36}$ (B) $a + b + c = -1$ (C) $\ell + m + n = \frac{19}{72}$ (D) $\lim_{x \rightarrow 4^+} f(x) = 0$
5. Let 'a' denotes the roots of equation $\cos(\cos^{-1}x) + \sin^{-1} \sin\left(\frac{1+x^2}{2}\right) = 2\sec^{-1}(\sec x)$ then possible values of $[|10a|]$ where $[.]$ denotes the greatest integer function will be
 (A) 2 (B) 5 (C) 10 (D) 1
6. An infinite geometric sequence is generated as $x, [x], \{x\}, \dots$ (where $[.]$ denotes greatest integer less than or equal to x and $\{x\} = x - [x]$ for $x \in \mathbb{R}$.)
 If sum of this sequence is expressed in the form $\frac{\sqrt{a} + b}{c - \sqrt{d}}$, where $a, b, c, d \in \mathbb{N}$ and are relatively prime numbers (a, c, d are prime numbers) then
 (A) $a + b + c + d = 14$
 (B) $a + b + c + d = 8$
 (C) function $f(x) = \frac{ax + b}{cx - d}$ is self invertible for $f : \mathbb{R} - \left\{\frac{d}{c}\right\} \rightarrow \mathbb{R} - \left\{\frac{a}{c}\right\}$
 (D) function $f(x) = \frac{ax + b}{cx - d}$ is not self invertible for $f : \mathbb{R} - \left\{\frac{d}{c}\right\} \rightarrow \mathbb{R} - \left\{\frac{a}{c}\right\}$

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **SIX** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+										
1	1	1	1	1	1	1	1	1	1	
2	2	2	2	2	2	2	2	2	2	
3	3	3	3	3	3	3	3	3	3	
4	4	4	4	4	4	4	4	4	4	
5	5	5	5	5	5	5	5	5	5	
6	6	6	6	6	6	6	6	6	6	
7	7	7	7	7	7	7	7	7	7	
8	8	8	8	8	8	8	8	8	8	
9	9	9	9	9	9	9	9	9	9	

1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If $f(x) = e^{[\cot x]}$, where $[y]$ denotes the greatest integer less than or equal to y , then the value

of $\frac{\lim_{x \rightarrow \frac{\pi^-}{2}} f(x)}{\lim_{x \rightarrow \frac{\pi^+}{2}} f(x)}$ is equal to (where $[.]$ denotes greatest integer function)

2. The number of integral values of x satisfying the equation $\operatorname{sgn}\left(\left[\frac{15}{1+x^2}\right]\right) = [1 + \{2x\}]$ is equal to (where $\operatorname{sgn}(x)$, $[x]$ and $\{x\}$ are signum of x , greatest integer function and fractional part function respectively)
3. Let $\mathbb{Z}$ be the set of integers, determine value of $f(10)$ where $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(2a) + 2f(b) = f(f(a+b)) \forall a, b \in \mathbb{Z}$ and $f(4) = 16$.

Space for Rough Work

4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a bijective function and the graph of $f(x)$ is symmetric about point $(1010, 0)$ then value of $f^{-1}(2020) + f^{-1}(-2020)$ is equal to
5. If $2 < x^2 < 3$ then the positive root of $\{x^2\} = \left\{\frac{1}{x}\right\}$, (where $\{x\}$ denotes the fractional part of x) is equal to (take $\sqrt{5} \approx 2.23$)
6. Let $\lim_{n \rightarrow \infty} \left(\sqrt[3]{3n^3 - 4n^2 + 5n + 1} - \lambda \sqrt[3]{81n^3 + 2n^2 + 1} \right) = \mu$ where μ & λ are finite numbers.

The value of $\left| \frac{\mu}{\lambda^{14/3}} \right|$ is equal to

Space for Rough Work

SECTION-III : (Maximum Marks : 18)

- This section contains **SIX** questions.
- The answer to each question is a **SINGLE DIGIT INTEGER** ranging from 0 to 9, both inclusive.
- For each question, darken the bubble corresponding to the correct integer in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. The value of $\left[x^2 \left(\frac{\cos(\cot^{-1} x) + \frac{1}{x} (\sin(\cot^{-1} x))}{\cot(\operatorname{cosec}^{-1} x) + \tan(\operatorname{cosec}^{-1} x)} \right)^2 + \frac{1}{x^4} \right]^{\frac{1}{2}}$ is equal to

2. The value of sum $\frac{2}{\pi} \left(\sum_{n=1}^{\infty} \tan^{-1} \left(\frac{8n}{n^4 - 2n^2 + 5} \right) + \tan^{-1} \frac{1}{2} \right)$ is equal to

Space for Rough Work

3. The number of solutions of the equation $1 + x^2 + 2x \sin(\cos^{-1}y) = 0$ is equal to 'k' and ordered pairs (x,y) which satisfy the equation are (a_i, b_i) for $i = 1, 2, 3, \dots, k$ then value of $\left(k + \sum_{i=1}^k a_i + \sum_{i=1}^k b_i \right)$ is
4. If $a, b, c \in \mathbb{R}^+$ and $a^2bc + 4abc + 4bc = 81$. The value of $a + b + c$ such that $2a + b + c + 4$ assumes its least value is equal to
5. Number of real solutions of the equation $2(2^x - 1)x^2 + (2^{x^2} - 2)x = 2^{x+1} - 2$ is equal to
6. The value of $\lim_{x \rightarrow \infty} \frac{(\sin x - 1)}{e^x}$ is equal to

Space for Rough Work

Space for Rough Work

QUESTION PAPER FORMAT AND MARKING SCHEME :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

17. Each part has three sections as detailed in the following table.

Section	Que. Type	No. of Que.	Category-wise Marks for Each Question				Maximum Marks of the section
			Full Marks	Partial Marks	Zero Marks	Negative Marks	
I	One or more correct option(s)	6	+4 If only the bubble(s) corresponding to all the correct option(s) is(are) darkened	+1 For darkening a bubble corresponding to each correct option, provided NO incorrect option darkened	0 If none of the bubbles is darkened	-2 In all other cases	24
II	Numerical Value Type (Up to second decimal place)	6	+4 If only the bubble corresponding to correct answer is darkened	—	0 In all other cases	—	24
III	Single digit Integer (0-9)	6	+3 If only the bubble corresponding to correct answer is darkened	—	0 If none of the bubbles is darkened	-1 In all other cases	18

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form number of the candidate, and that question paper and ORS codes are the same.

Signature of the Invigilator